A deep learning approach for Control Chart Patterns (CCPs) prediction

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Outline

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Introduction

Context:

- In manufacturing, a control chart is used to monitor product quality or process characteristics.
- Any failure in the manufacturing process would cause abnormal patterns in the control chart.

Motivation:

- There is a lot of research focused on pattern recognition or classification related to control charts for detection:
  - Using the neural network for online – control chart pattern detection and discrimination [1]
  - Feature-based recognition of control chart patterns [2]
  - …
- In this study, we propose another approach to control chart patterns which is prediction.

Problem description

Assumption

- Time series data is consist of 6 popular control chart patterns (Figure 1).

- There is a degradation of process performance during operation

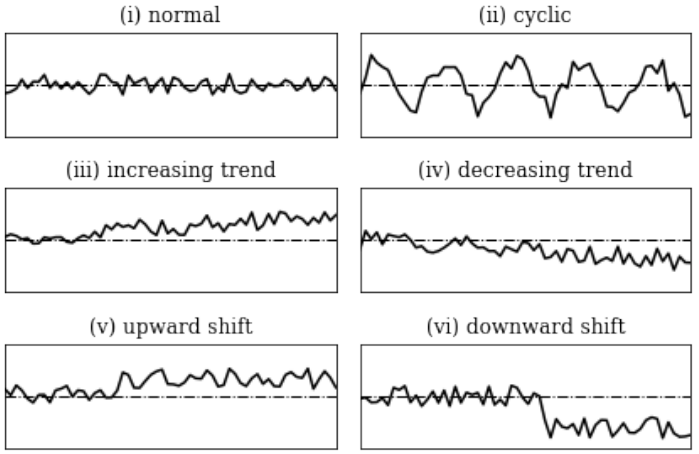


Figure 1. Control chart patterns

Description

- Given the historical data of a time series with a length of n x T, I.e., $X = (p_1, \ldots, p_T)$ in which $p_i \in R^n$ is a control chart pattern with n samples. We aim to achieve goals:

  With a fixed length view of past n_I values, a predictor f aims at predicting the next n_O values.

Proposed method

In the past, given this problem there are several methods are used:

- Autoregressive moving average models (ARMA) [3]
- Autoregressive-integrated moving average (ARIMA) and seasonal ARIMA models [4]
- …

⇒ Linearity assumption for the system being observed.

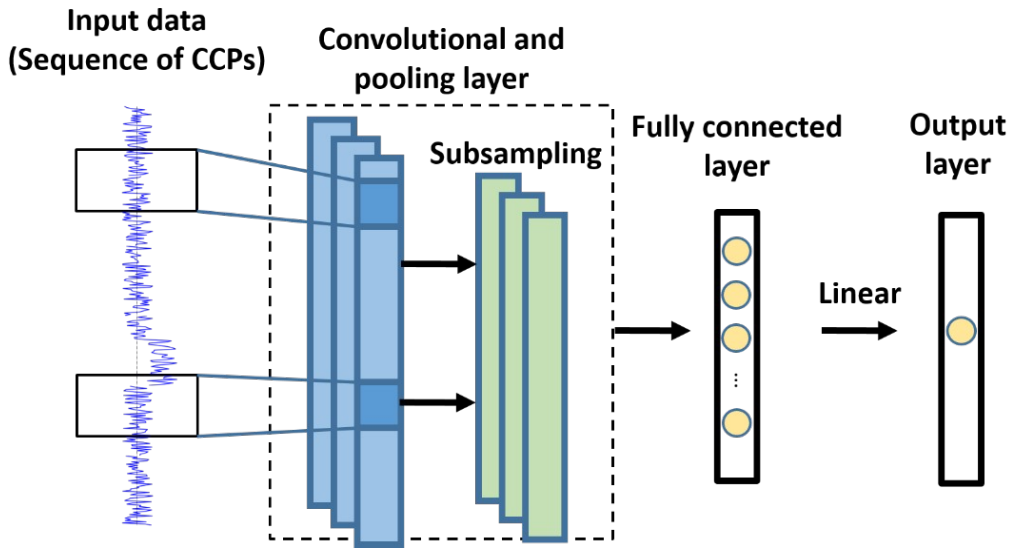
- Kernel-based methods and support vector machine (SVM) [5]

⇒ Choosing kernel or feature map manually.

Non-linear models like **convolution neural networks** were proposed to overcome these problems.

Proposed method

Convolutional neural network (CNN) LeCun et al. (1995),



- The convolutional layer is to learn and extract feature maps from data.
- The fully connected layer has the ability to connect these features and then predict the next values of the time series.

Loss function

The mean squared error:

$$L(\theta) = \frac{1}{N}\sum_{t=1}^{N}\left(y_t - f(p_t;\theta)\right)^2$$

N: number of samples.

y_t: real output of time series.

$f(p_t;\theta)$: output of CNN model, in which p_t is the input vector, θ is the network's parameters.

Performance assessment

The efficiency of the model has been measured and quantified by:

- Root mean squared error (RMSE)
- Mean absolute error
- Normalised root mean squared error (%)
- R2 score

Data

To our knowledge, the time series data of the control chart pattern is not available at this moment. We decided to use simulated data (Figure 2).

- First, we create control chart patterns (CCPs) with *n* samples (based on the equations from Western Electric Company (1958))
- After that, we connect these CCPs into time series based on their performance index C_p:

$$C_p = \frac{USL - LSL}{6\hat{\sigma}}$$

$$\hat{\sigma} = \frac{\max(x_i(t)) - \min(x_i(t))}{d_2}, t \in [0, T)$$

With d_2 is the statistical constraint which depends on the sample size, T is the window length of CCPs, $\hat{\sigma}$ is the estimated standard deviation of CCP and the upper or lower specification limit is *USL, LSL* respectively.

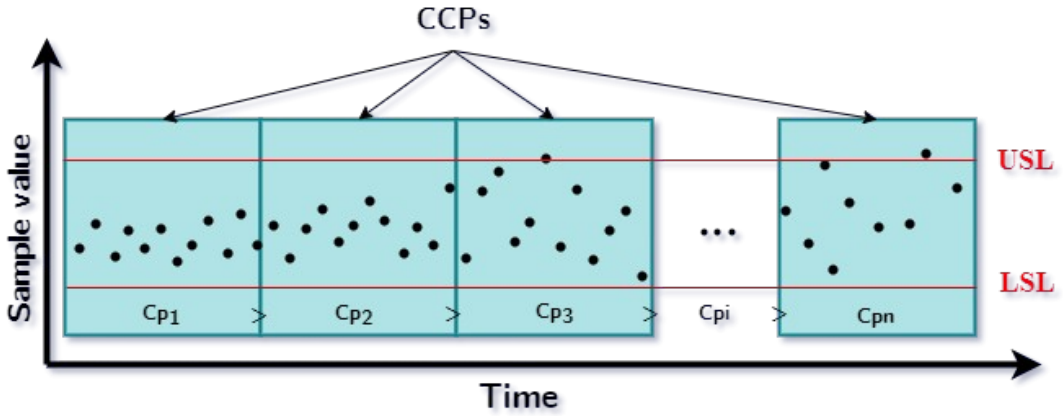
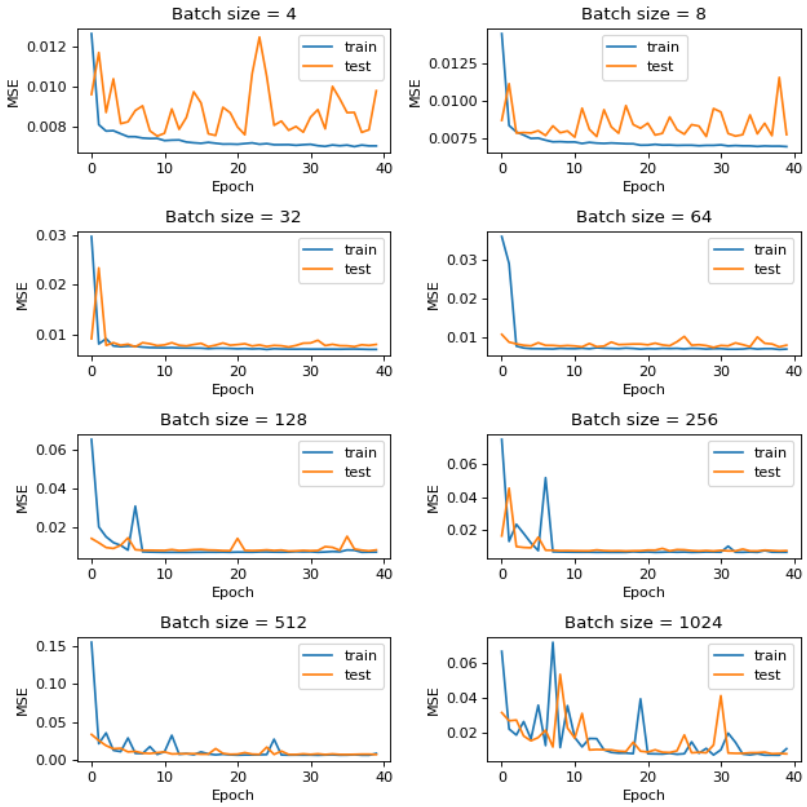
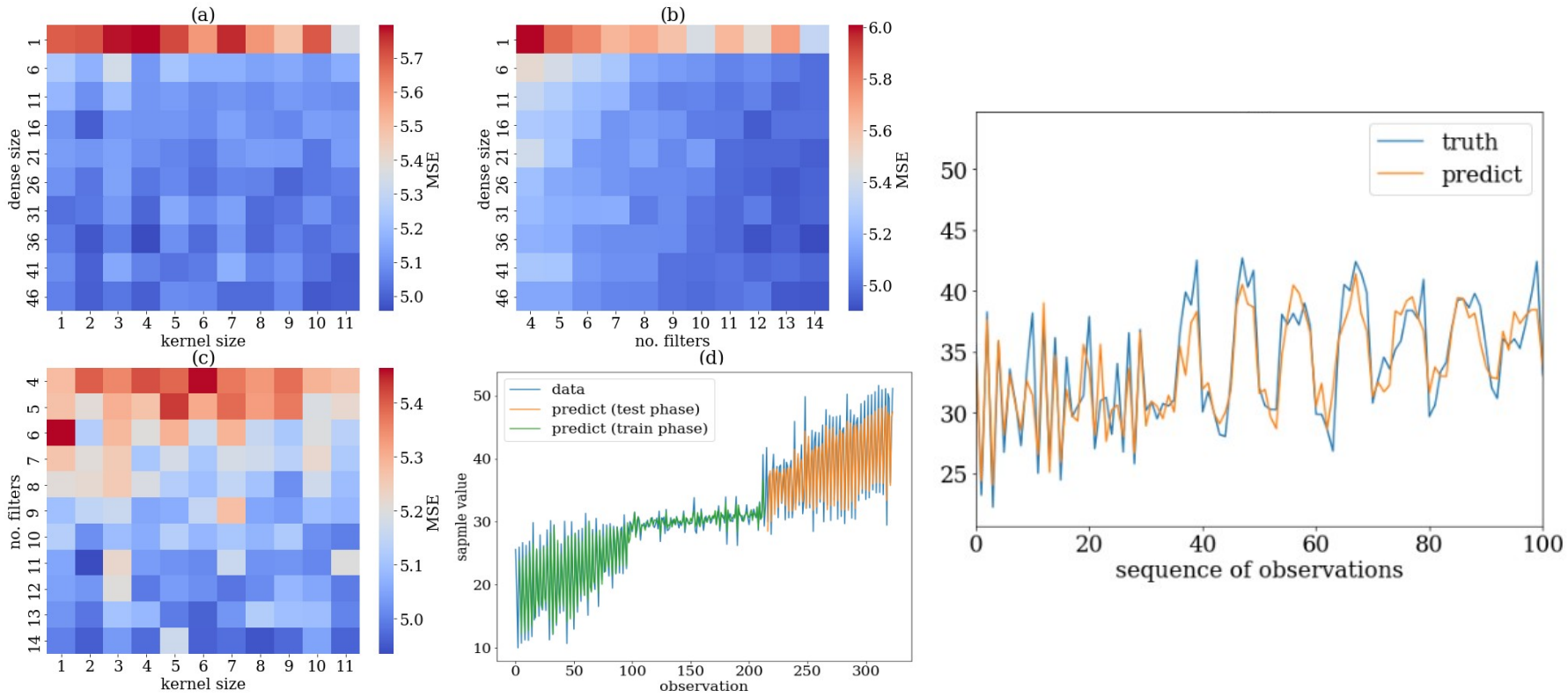


Figure 2. The time series of CCPs.

The best results were obtained with a batch-size b = 32 and epochs e = 40 in which Nadam (Dozat, 2016) is used as optimizer.



The influence of parameter to Model's performance



Heatmaps (a)-(c) show the MSE of the prediction of mixed patterns. The values are mean values across the third parameter. In which: (a) shows the MSE dependent on the dense size and the kernel size. (b) shows the MSE dense size and the number of filters. (c) shows MSE dependent on the kernel size and the number of filters and (d) shows a prediction using the trained CNN.

Cross-validation

Dataset	Best model configuration (model in this study)	Model performance			
		RMSE	MAE	NRMSE%	R2
Dataset (1): mixed pattern	dens size = 46	5.75	4.58	16.43%	0.42
	number of filter = 14	5.71	4.52	16.3%	0.41
	kernel size = 11	5.58	4.31	15.93%	0.34
	input length = 120	6.47	5.16	18.48%	0.36
		5.53	4.39	15.79%	0.28
dataset (2) normal patterns	dense size = 46	3.82	3.16	31.75%	-5.92
	number of filter = 14	3.81	3.16	31.67%	-8.03
	kernel size = 10	3.65	3.1	30.44%	-6.22
	input length = 120	3.56	3.06	29.67%	-15.82
		3.57	3.07	29.93%	-11.67

Table 1.: The performance of the best model configuration (in all models configurations are investigated in this study)

Conclusions and perspectives

- Predicting CCPs of the manufacturing process is possible with relatively simple CNN when the parameters are chosen correctly.

- The performance of the CNN model is still limited and should be improved with **other deep learning methods**.

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Thank for listening